

Loc Nguyen

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SUMMARY

Data Analytics graduate with hands-on experience developing, testing, and deploying AI-enabled solutions across the full machine learning lifecycle. Skilled across Microsoft Office suite (Excel, Word, Outlook), Python, SQL, with applied expertise in data modelling, computer vision, and anomaly detection to solve real-world business problems. Developed a DETR-family model that generated approximately \$2.64M in potential infringement value and automated ML pipelines to cut manual intervention by 70%. Brings a curious, proactive approach to continuous improvement and responsible AI practice, ready to deliver real impact through [Company].

EXPERIENCE

ONETASK

Data Analytics Intern

Docklands, VIC

July 2025 – November 2025

- Developed and optimised an RF-DETR model achieving **88% mAP** on NVIDIA Jetson Nano for automated phone-use detection, delivering an **8%** performance uplift over the prior YOLOv4 solution and enabling identification of **4,218** potential offenses, representing approximately **\$2.64M** in **potential infringement value**.
- Automated end-to-end machine learning workflows via Apache Airflow, reducing manual pipeline intervention by **70%** and improving dataset reproducibility across training runs.
- Built and presented an interactive Power BI dashboard consolidating benchmarked scores of three computer vision models (YOLOv4, YOLO-NAS, RF-DETR) and **20+ experiments** to business managers, profiling cost-versus-performance trade-offs and pricing implications to support data-driven decisions.

CONSTELLATION TECHNOLOGIES LIMITED

Data Analytics Intern

Melbourne, VIC

July 2024 – November 2024

- Designed a BERT-based text-extraction system achieving **95% extraction accuracy** across product data from **79 suppliers**, establishing the data-processing foundation for an IT-developed application that accelerated customer access to product details.
- Built an Isolation Forest anomaly-detection model to flag irregular temperature patterns with **99% accuracy**, supporting earlier identification of quality risks across supply chain data.
- Enhanced Power BI dashboards with root-cause analysis to identify key outlier factors driving redundant goods costs and product-condition issues, uncovering **12%** cost variance and directly informing pricing information during peak periods.

ONETASK

Data Specialist

Docklands, VIC

March 2023 – Present

- Run daily QA reviews across **3,500+** AI-processed records, flagging data discrepancies and feeding accuracy statistics into senior stakeholder reporting.
- Conduct monthly data quality audits in Excel (VLOOKUP, Pivot tables) using structured templates to validate system outputs, reducing misclassification rate by **9%** over three months.
- Created and maintain **Excel** visualisations of offence trends by camera location, communicating high-volume hotspot patterns of **300+** daily detections to support enforcement prioritisation decisions and resource-allocation decisions.
- Act as liaison between business and engineering teams, resolving **7+** recurring data-quality issues and coordinating resolution with cross-functional stakeholders, cutting downstream data loss via system calibration.

EDUCATION

Monash University

Bachelor of Applied Data Analytics Advanced (Honours)

First Class Honours

Honours Thesis: Application of Detection Transformers in Live Traffic Monitoring Systems

Clayton, Australia

December 2025

PROJECTS

EUROPEAN BANK CHURN ANALYSIS

- Built RFM segmentation logic in SQL from a **10,000-customer** dataset and developed Power BI dashboards surfacing churn risk and retention rate, with recommended actions per customer segment.
- Managed experiment tracking (MLflow) and hyperparameter optimisation (Optuna), deploying the best XGBoost model with **72% accuracy** via Streamlit for real-time churn predictions.

SALES FORECASTING PROJECT (Unit Capstone Group Project)

- Led a five-member team through full data lifecycle delivery with weekly team meetings and supervisor updates, feature engineering variables to analyse store sales with Python (Pandas), applying K-means customer sales segmentation (**98% accuracy**) and Random Forest demand forecasting (**95% accuracy** in 4-week sales horizon).

TECHNICAL SKILLS

Data & Office Tools: Python (Pandas, Numpy, Matplotlib, Scikit-Learn, Pytorch), SQL, R, Microsoft 365 products (Excel, Word, Outlook), Power BI (DAX, Power Query)

Machine Learning: Linear/Logistic Regression, Decision Trees, Random Forest, KNN, K-means, Anomaly Detection, Regularisation Techniques (Ridge, Lasso), NLP, Computer Vision

Pipelines: Apache Airflow, MLflow, Optuna, Git, Linux (Ubuntu), Docker, Streamlit

Core Competencies: AI Solution Development, Root-Cause Analysis, Data Quality Auditing & Governance, Cross-Functional Collaboration, Process Automation